FPGA Hardware Implementation of ECG Waveform Segmentation

Panagiotis Zintzovas - RN: 1096767

University of Patras,
MSc «Hardware and Software Integrated Systems»
July 2025

FPGA Hardware Implementation of waveform segmentation of human ECG signals using LSTM network

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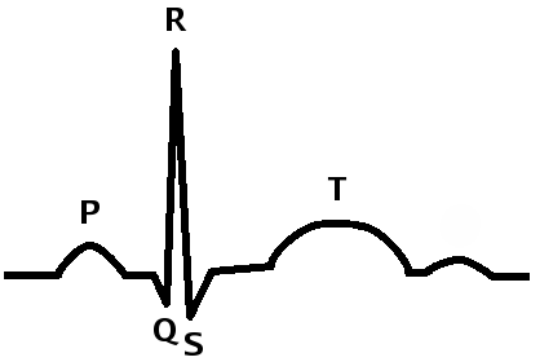
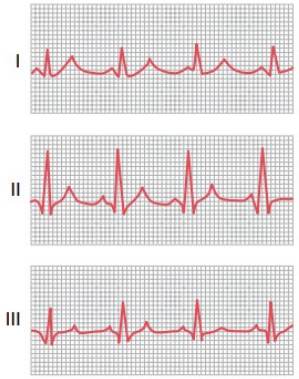
Outline

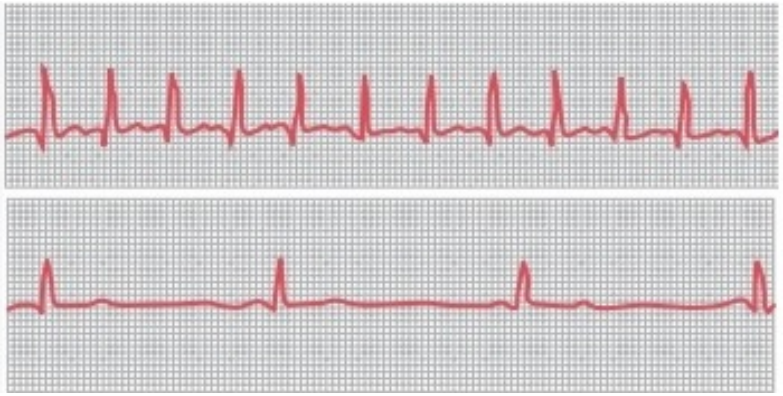
1. ECG Introduction
2. LSTM Networks: Overview
3. MATLAB rawNet
4. C++ Forward Pass
5. Quantization-Aware Training
6. HLS & RTL Implementation
7. Results & Discussion

What Is an ECG?

- An electrocardiogram (ECG) records the heart's electrical activity over time.
- Key waveform components:
  - **P wave:** atrial depolarization
  - **QRS complex:** ventricular depolarization
  - **T wave:** ventricular repolarization
- Timing and morphology of these waves (e.g. QRS duration) are critical for diagnosing arrhythmias, infarction, etc.

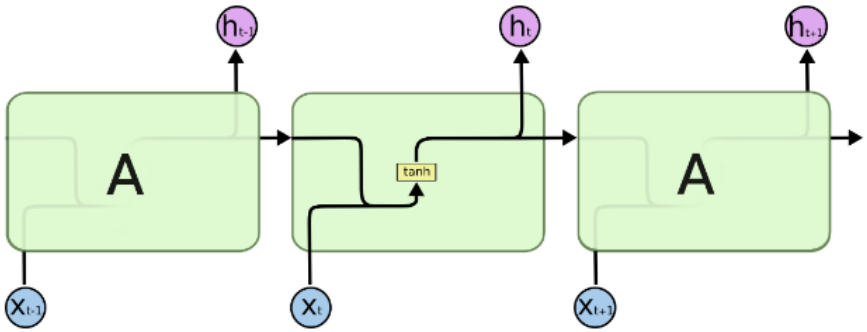




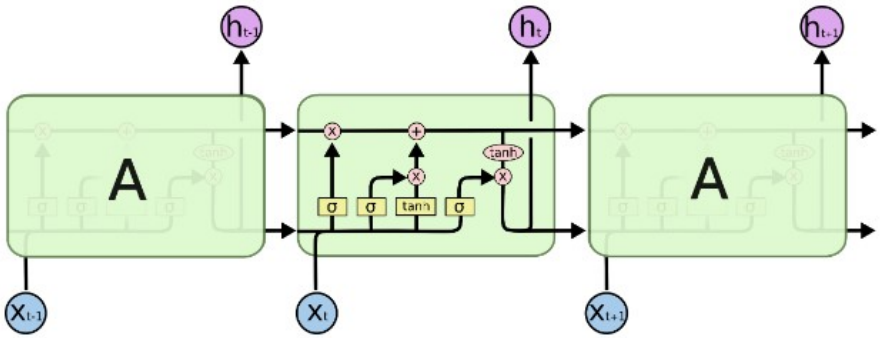


LSTM Networks: Overview

- LSTM = Long Short-Term Memory
  - A type of RNN designed to learn long-range dependencies
  - Avoids vanishing/exploding gradients via gated cells
- Standard RNN "repeating module" vs. LSTM cell:



Standard RNN cell



LSTM cell (4 gates)

- LSTM gates at time t: i_t (input), f_t (forget), o_t (output), $\tilde{c}_t$ (cell candidate)

rawNet: Data, Architecture & Training

- **Raw input:** 1D ECG segments (5000 samples) → 2835 segments from 210 recordings
- **Tall arrays (in-memory):**

```
trainData = gather(tall(trainDs));
testData  = gather(tall(testDs));
```

- **Network & Training:**

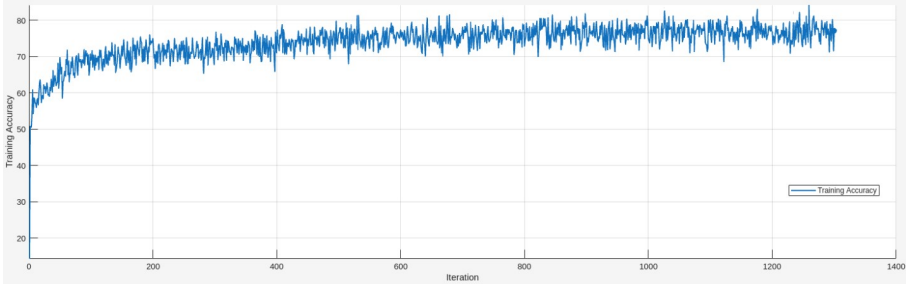
Layers:

```
layers = [ ...
  sequenceInputLayer(1)
  lstmLayer(200,'OutputMode','sequence')
  fullyConnectedLayer(4)
  softmaxLayer];
```

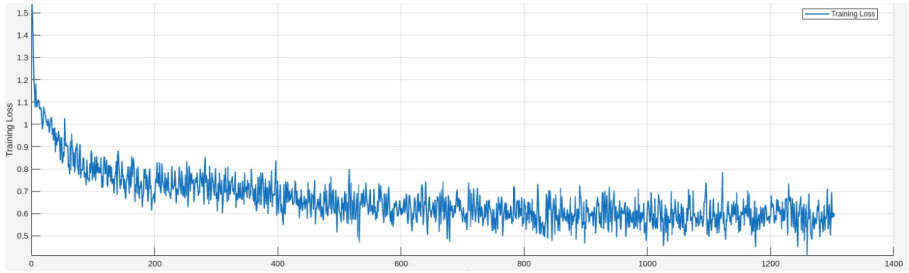
Options:

```
options = trainingOptions('adam', ...
  'InitialLearnRate',0.01, ...
  'MaxEpochs',10, ...
  'MiniBatchSize',50, ...
  'LearnRateSchedule','piecewise', ...
  'LearnRateDropFactor',0.1, ...
  'LearnRateDropPeriod',3, ...
  'GradientThreshold',1);
```

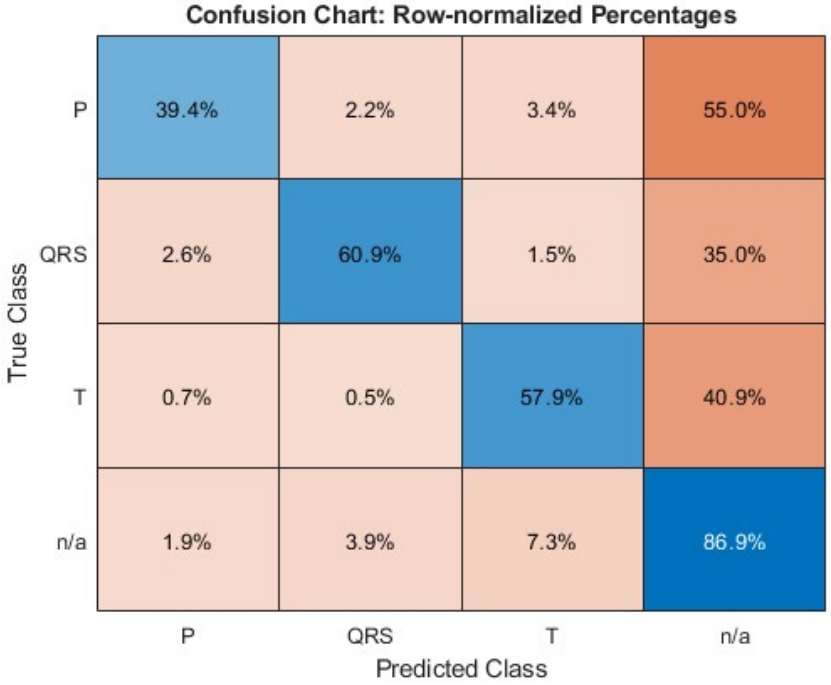
Training & Test Results



Training accuracy (10 epochs)



Training loss

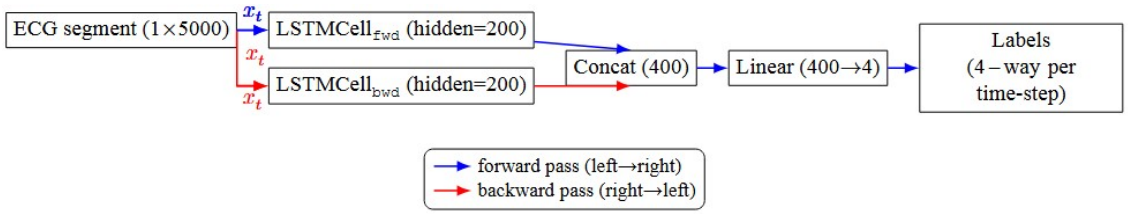


Row-normalized confusion chart on test set

- Per-class test accuracies: n/a 86.9%, P 39.4%, QRS 60.9%, T 57.9%.

C++ Forward-Pass: Design Highlights

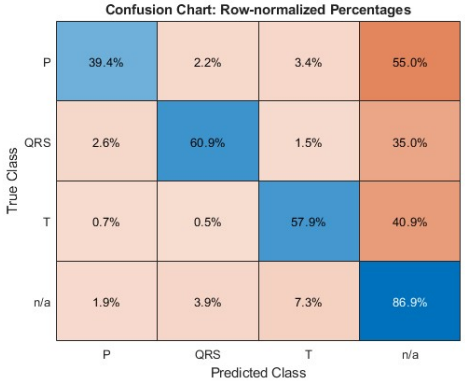
- **One-to-one functional mapping** We took MATLAB's rawNet LSTMCell+FC and rewrote it in plain C++ – same I–F–O–G gate order, same sequence unroll – "ping-pong" double-buffering to mirror hardware streaming
- **Lightweight dependencies** Uses only <vector>, <algorithm>, and <cmath>
- **Chunked processing** Reset hidden/cell state per 5000-sample segment → exactly MATLAB semantics

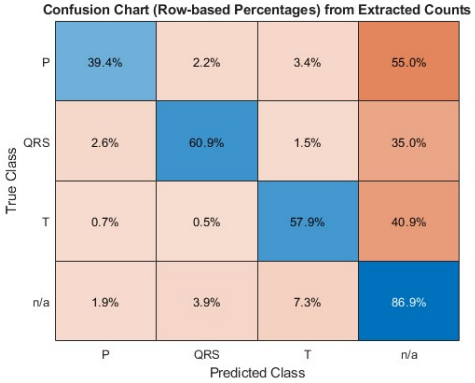


Conceptual data-flow: per-sample cell → softmax → argmax

C++ Forward-Pass: Validation & HLS Prep

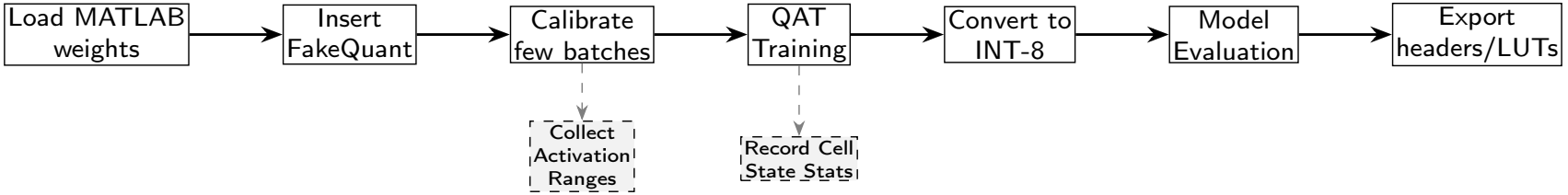
- **Correctness** → C++ vs. MATLAB confusion matrix matches within floating-point tolerance
- **Performance** – Single-threaded CPU: X ms per 5000-sample segment – Leaves plenty of margin for hardware acceleration
- **Next step: HLS** – Quantized weights & activations → ap_fixed<W,I> headers – LUT-based sigmoid/tanh → read-only tables – Automatic translation of our C++ forward-pass into Vivado HLS





QAT: Motivation & Workflow

- **Why QAT?**
  FPGA DSPs are resource-hungry for FP32. QAT simulates 8-bit effects during training to retain accuracy.
- **Overall Pipeline**
  1. Insert *FakeQuant* observers in PyTorch FX
  2. *Calibrate* on a few batches → collect scales
  3. *Train* for 30 epochs with weighted sampling
  4. *Convert* to INT-8 and export headers/LUTs



Key QAT Details

FakeQuant Configs

- `W_FQ`: symmetric per-channel ($\pm 127 \rightarrow$ qint8)
- `A_FQ`: affine per-tensor ($0 \ldots 255 \rightarrow$ quint8)

Class Balancing

$$w_i = 1 + 2\,(1 - f_{n/a}^{(i)})$$

WeightedRandomSampler oversamples P/QRS/T-rich segments.

Training Schedule

- 30 epochs, Adam @ 1e-4
- LR $\times 0.3$ at epoch 15
- 20-batch calibration before epoch 1

Hooks & Stats

- Capture gate max $g_{\max}$
- Track cell-state μ/σ & percentiles
- Suggest `ap_fixed<16,6>` for `state_t`

QAT Export & Results

HLS-Ready Artifacts

- *Weight & bias headers* (ap_fixed<16,6>): `W_all_fixed.h`, `B_all_fixed.h`, ...
- *Activation LUTs* (Q2.6): `sigmoid_lut.h`, `tanh_lut.h`, `exp_lut.h`
- *Test vectors*: `ecg_testData_int8.bin`
- *Stats & suggestion*: `act_ranges.json`, `qat_stats.json`, `datatype_suggestion.json`

Per-class Accuracy (INT-8 vs MATLAB)

Class	MATLAB %	INT-8 %	Δ (pp)
n/a	86.9%	86.68%	−0.22%
P	39.4%	40.25%	+0.85%
QRS	60.9%	61.00%	+0.10%
T	57.9%	58.38%	+0.48%

Compared to the MATLAB baseline, QAT yields a slight dip in "n/a" but measurable gains in all true-event classes, setting us up for a fully deterministic HLS flow.

Fixed-Point Data Types

Type	Purpose / Range
`input_t`	`ap_int<8>` 8-bit signed ECG samples
`state_t`	`ap_fixed<16,6>` hidden/cell (10 frac bits)
`acc_t`	`ap_fixed<32,24>` MAC accumulator (200 taps)
`cell_t`	`ap_fixed<26,14>` extra headroom for cell update
`sum_t`	`ap_fixed<24,6>` softmax sum/exponentials
`in_pkt_t`/ `out_pkt_t`	`ap_axiu<8,0,0,0>` AXI-Stream word

Why these widths? Derived from our QAT-hook statistics (μ/σ, $g_{\max}$, weight sums) to avoid overflow and minimize area.

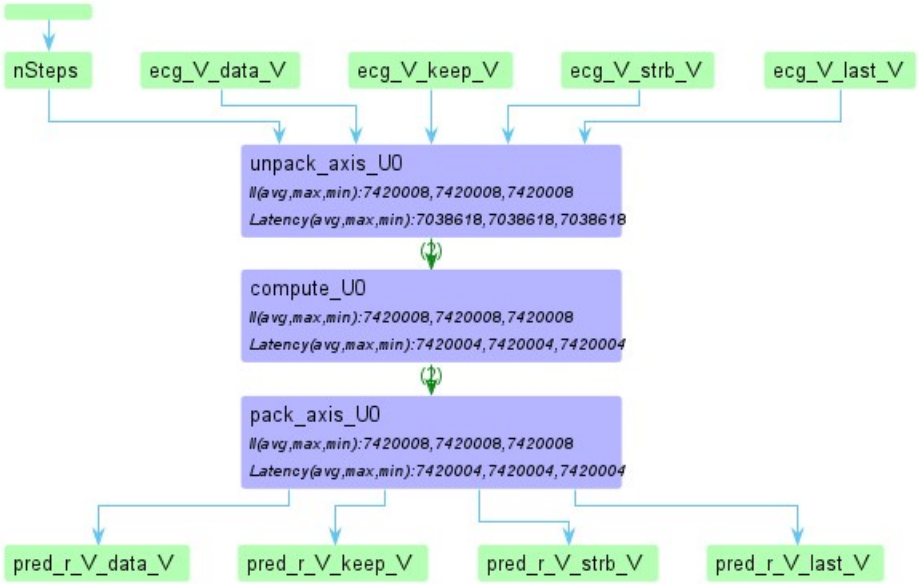
HLS Kernel & Key Pragmas

```
/* lstm_forward.cpp */
#pragma HLS INTERFACE axis   port=input_stream
#pragma HLS INTERFACE axis   port=output_stream
#pragma HLS INTERFACE s_axilite port=return

#pragma HLS DATAFLOW
void lstm_forward_chunk(...) {
  #pragma HLS PIPELINE II=1
  for(int t=0; t<chunkSize; ++t) {
    lstm_cell(...);
    softmax(...);
  }
}
```

- `INTERFACE axis` – AXI-Stream I/O ports
- `INTERFACE s_axilite` – Control interface bundle
- `BIND_STORAGE` – Place weight arrays in BRAM
- `ARRAY_PARTITION` – on LUTs & state buffers for single-cycle access
- `DATAFLOW` – overlap read/compute/write in chunk wrapper

Dataflow Concurrency & Schedule



HLS Estimates vs. Co-Sim & Resources

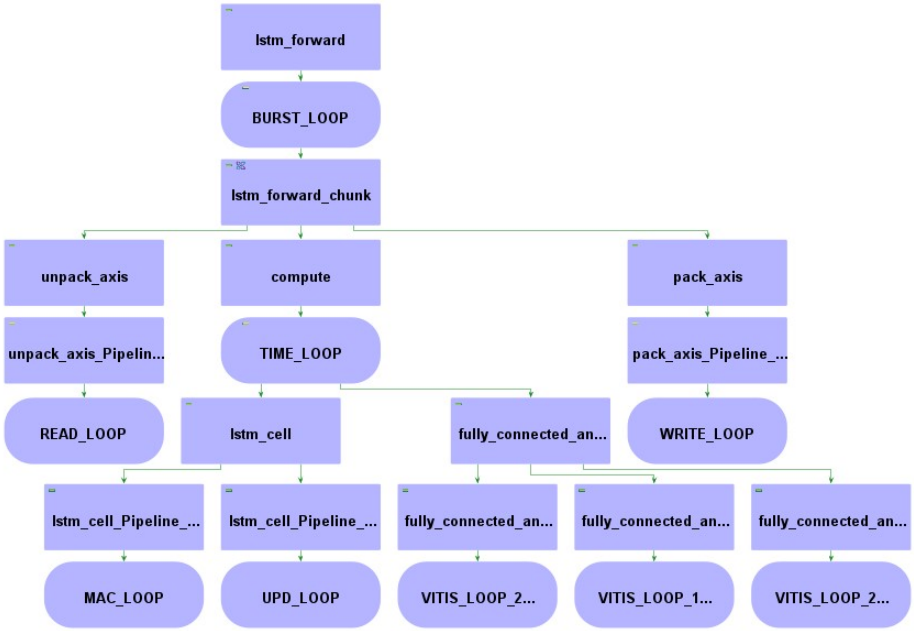


Figure: Function/loop hierarchy

Clock & Latency

	HLS	RTL
Clock period (ns)	6.108	6.256
$f_{\max}$ (MHz)	163	160
II	1	1
Latency/sample (cyc)	1484	1502

Resources

Resource	Used	Util %
BRAM_18K	410/624	65
DSPs	410/1728	23
FFs	47 249/460 800	10
LUTs	54 284/230 400	23

Vivado Implementation

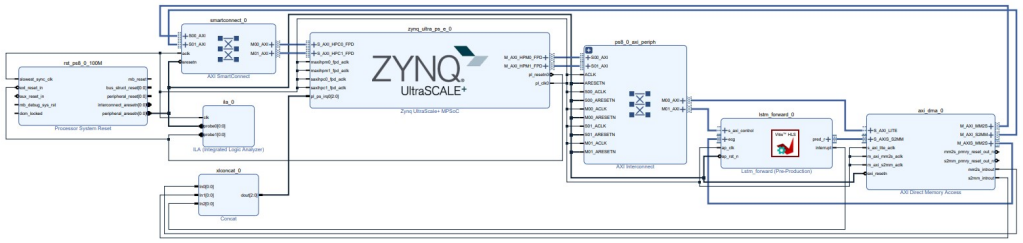


Figure: Vivado block design

Resource Utilization

Resource	Used	Util %
CLB LUTs	18 640	8.1
Registers	50 481	11.0
Block RAM	208.5	66.8
DSP48E2	410	23.7

Power Breakdown

Component	Power (W)
Total Dynamic	3.369
Total Static	0.696

Timing Summary

- PL clock = 100 MHz
- Worst Negative Slack: 1.403 ns
- Longest path: 8.295 ns

FPGA LSTM Forward Pass: Host-Side Loop

- Bitstream on ZCU104 → bare-metal A53 app
- Streams 5 000-sample bursts via AXI-DMA
- Measures wall-clock via ARM timer
- Prints per-class confusion matrix

```
// init HLS core
XLstm_forward core;
XLstm_forward_Initialize(&core, XPAR_XLSTM_FORWARD_0_DEVICE_ID);
// pointers ...
auto ecg = (volatile uint8_t*) ECG_BASE_PHYS;
auto pr  = (volatile uint8_t*) PR_BASE_PHYS;
// start timer
XTime t0,t1; XTime_GetTime(&t0);
// stream bursts
uint32_t done=0;
while(done<TOTAL_SAMPLES){
  uint32_t n = std::min(CHUNK_SAMPLES,TOTAL_SAMPLES-done);
  XLstm_forward_Set_numSamples(&core,n);
  XLstm_forward_Set_chunkSize(&core,n);
  XLstm_forward_Start(&core);
  dma_mm2s((uintptr_t)&ecg[done], n);
  dma_s2mm((uintptr_t)&pr [done], n);
  wait_core_idle(&core);
  done += n;
}
// stop & report
XTime_GetTime(&t1);
double secs = double(t1-t0)/double(CPU_FREQ);
printf("Total inference time: %.3f s\n",secs);
// confusion
print_confusion(gt,pr,TOTAL_SAMPLES);
```

FPGA Throughput & Real-Time Speed-up

Table: FPGA LSTM Forward-Pass Throughput

Total samples processed	14 175 000
Total inference time	210.373 s
Sustained throughput	$\frac{14.175 \times 10^6}{210.373} \approx 6.74 \times 10^4$ samples/s
Per-segment latency (5 000)	$\frac{210.373}{2835} \approx 0.0742$ s
Real-time factor	$\frac{20\,\mathrm{s}}{0.0742\,\mathrm{s}} \approx 270\times$

These include all ARM↔FPGA transfers, kernel execution, and result fetch—demonstrating over two orders of magnitude beyond real-time.

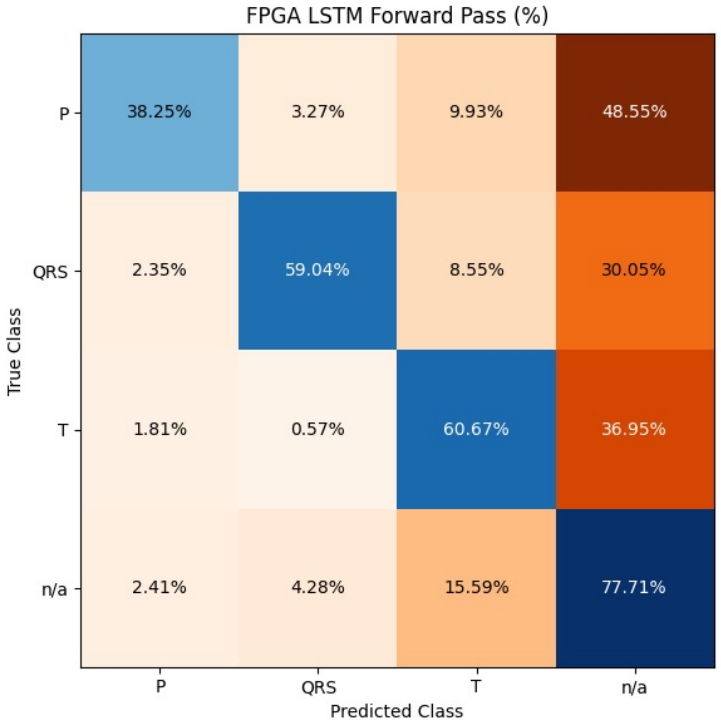
FPGA Per-Class Accuracy

Confusion Matrix (row-norm.%)

	P	QRS	T	n/a
P	38.25%	3.27%	9.93%	48.55%
QRS	2.35%	59.04%	8.55%	30.05%
T	1.81%	0.57%	60.67%	36.95%
n/a	2.41%	4.28%	15.59%	77.71%

Per-Class Accuracy

Class	Accuracy	Correct/Total
P	38.25%	589 618/1 541 322
QRS	59.04%	1 215 485/2 058 757
T	60.67%	2 258 573/3 722 547
n/a	77.71%	5 325 013/6 852 374



Row-norm. confusion chart

Cross-Platform Accuracy Comparison

Class	MATLAB FP	C++ FP	QAT INT-8	FPGA	$\Delta_{\text{FPGA-MATLAB}}$
n/a	86.9%	86.9%	86.68%	77.71%	-9.19 pp
P	39.4%	39.4%	40.25%	38.25%	-1.15 pp
QRS	60.9%	60.9%	61.00%	59.04%	-1.86 pp
T	57.9%	57.9%	58.38%	60.67%	+2.77 pp

Key take-aways:

- FPGA drops 1–2 pp on P/QRS but gains on T.
- Large "n/a" drop driven by QAT epochs (30 vs 10) & weighted sampling.
- Hardware sustains $\approx 270\times$ real-time with 23
- Future: mixed-precision, pruning, multi-lead ECG.

Any Questions?

Thank You!

Panagiotis Zintzovas - phy5286@ac.upatras.gr / st1096767@ceid.upatras.gr